
SIMATS ENGINEERING

Department of [AI & DS]

CASE STUDY ANALYSIS

AI-Powered Smart Volunteer and NGO Resource Management Platform

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1. Introduction and Industry Context

Non-Governmental Organizations (NGOs) and grassroots volunteer groups form the backbone of humanitarian service delivery across education, healthcare, disaster relief, and community development sectors. Their impact depends heavily on the efficient coordination of volunteers, timely allocation of resources, and transparent tracking of donations. As the scale of NGO operations grows, traditional manual coordination methods — spreadsheets, phone calls, and disconnected paperwork — become increasingly inadequate, leading to delayed responses, volunteer disengagement, and inefficient use of scarce resources.

The convergence of Artificial Intelligence (AI), cloud computing, Geographic Information Systems (GIS), and mobile technologies presents a significant opportunity to modernize NGO operations. An intelligent, unified digital platform can automate volunteer-task matching, provide real-time visibility into project progress, and build donor confidence through transparent resource tracking — ultimately maximizing the social impact that NGOs are able to deliver.

Problem Statement

NGOs often face difficulties in managing volunteers, assigning tasks, tracking donations, and monitoring project outcomes due to fragmented systems. Manual coordination leads to delays and inefficient resource utilization. The proposed AI-Powered Smart Volunteer and NGO Resource Management Platform matches volunteers to suitable activities using AI, optimizes resource allocation, monitors project progress, tracks donations, and provides real-time dashboards for administrators — thereby improving transparency, volunteer engagement, and operational efficiency across the organization.

Objectives

1. Manage volunteer registrations digitally through a centralized platform.
2. Match volunteers to suitable activities using AI-driven recommendation techniques.
3. Monitor NGO project progress in real time across all active initiatives.
4. Track donations and resources with complete transparency and accountability.
5. Optimize volunteer allocation to maximize utilization and minimize idle capacity.
6. Generate organizational reports for internal decision-making and external stakeholders.
7. Improve stakeholder communication between volunteers, coordinators, and donors.
8. Enhance operational transparency across all NGO activities and financial transactions.

Expected Outcomes

1. Improved volunteer participation and retention.
2. Efficient allocation of human and material resources.

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3. Better project management with reduced delays.
 4. Enhanced organizational transparency and accountability.
 5. Increased donor confidence through visible impact tracking.
 6. Reduced administrative workload for NGO staff.
 7. Improved quality and reach of community services.
 8. Greater overall social impact per unit of resource invested.

2. Project Vision and Stakeholders

Project Vision

To build an intelligent, unified digital ecosystem that transforms fragmented and manual NGO operations into a data-driven, transparent, and efficient platform — where volunteers are matched to the causes best suited to their skills, NGOs can track resources and donations in real time, and administrators are empowered with actionable, AI-driven dashboards that maximize social impact for the communities they serve.

Alignment with Objectives

The platform's vision directly supports the stated objectives by digitizing volunteer onboarding, embedding AI-based matching at the core of task allocation, and layering real-time monitoring and reporting over every project and donation. This creates a single source of truth that reduces administrative overhead while increasing transparency for every stakeholder involved in the NGO's mission.

Key Stakeholders

The table below identifies all key stakeholders involved in the platform, along with their primary role and interest in the system.

Stakeholder	Role / Description	Primary Interest
Volunteers	Individuals who register, get matched to tasks, and execute on-ground activities.	Meaningful, well-matched opportunities and recognition
NGO Administrators / Program Managers	Oversee projects, approve registrations, monitor progress, and generate reports.	Operational efficiency and organizational oversight
Field Coordinators / Team Leads	Supervise on-ground execution and assign day-to-day tasks to volunteers.	Real-time task tracking and volunteer accountability

Stakeholder	Role / Description	Primary Interest
Donors & Sponsors	Individuals or organizations contributing funds or resources.	Transparency and verifiable impact of contributions
Beneficiary Communities	Recipients of NGO programs and services.	Timely and quality delivery of services
Partner Organizations	Collaborating NGOs and CSR partners sharing resources or projects.	Seamless coordination and shared visibility
Government / Regulatory Bodies	Oversight bodies requiring compliance and audit reporting.	Regulatory compliance and fund-utilization transparency
System Administrators / IT Team	Maintain platform infrastructure, security, and uptime.	System reliability, security, and scalability

Figure 1: Stakeholder Ecosystem of the NGO Resource Management Platform

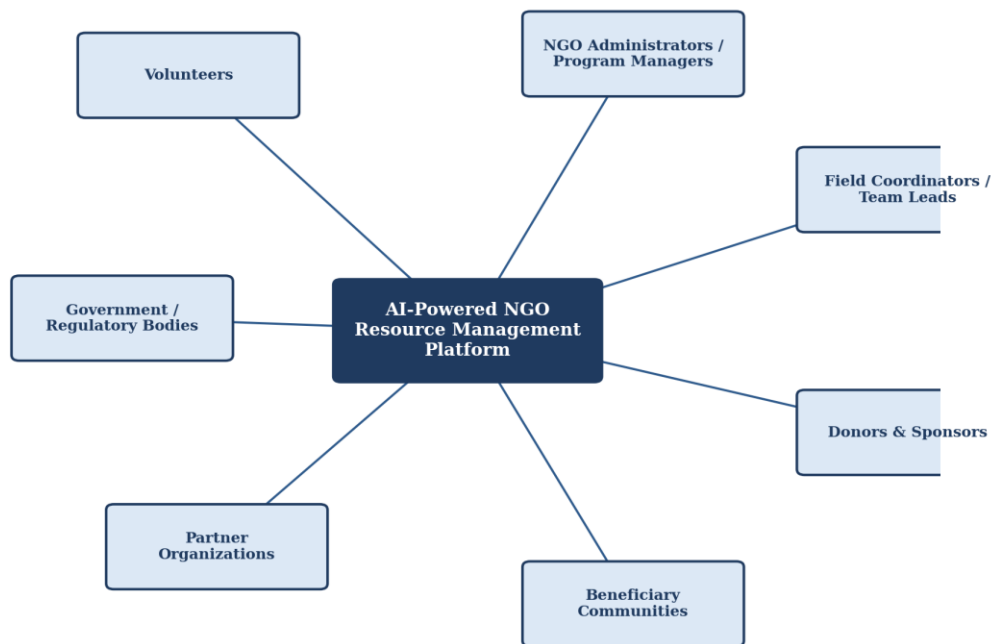


Figure 1: Stakeholder Ecosystem of the NGO Resource Management Platform

3. Epics and User Stories

The requirements of the platform were decomposed into seven Epics, each representing a major functional capability. Every Epic was further broken down into detailed User Stories written in the

standard Agile format: “As a <user>, I want <feature>, so that <benefit>.” This ensures each requirement is grounded in a clear user role, a concrete feature, and a tangible business benefit.

Epic Summary

Epic ID	Epic Name	Description
EPIC-1	Volunteer Management & Registration	Enables individuals to register as volunteers and build digital profiles capturing their skills, interests, availability, and location so that NGOs can identify suitable participants for their programs.
EPIC-2	AI-Powered Volunteer–Task Matching	Uses AI/ML algorithms to automatically match volunteers with the most suitable tasks based on skills, location, availability, and past performance, replacing slow and inefficient manual allocation.
EPIC-3	Project & Task Progress Monitoring	Provides tools to track the real-time status of NGO projects and individual tasks so that administrators can identify delays and intervene proactively.
EPIC-4	Donation & Resource Tracking	Allows donors to contribute funds or resources and enables NGOs to record and report how each donation is utilized, improving transparency and donor trust.
EPIC-5	Reporting & Analytics Dashboard	Generates automated organizational reports and visual analytics summarizing volunteer engagement, project outcomes, and resource utilization for administrators and regulators.
EPIC-6	Communication & Notification System	Facilitates real-time communication between volunteers, coordinators, and donors through notifications, alerts, and broadcast messaging to improve engagement.
EPIC-7	Transparency & Compliance Management	Ensures the platform maintains audit trails, data integrity, and regulatory compliance so that trust is preserved among all stakeholders.

Detailed User Stories by Epic

EPIC-1: Volunteer Management & Registration

- US-01 — As a volunteer, I want to create a digital profile with my skills, interests, and availability, so that NGOs can identify me for suitable activities.

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- US-02 — As a volunteer, I want to upload identity and certification documents, so that my profile is verified and trusted by NGOs.
 - US-03 — As an NGO administrator, I want to review and approve volunteer registrations, so that only verified volunteers are onboarded onto the platform.

EPIC-2: AI-Powered Volunteer–Task Matching

- US-04 — As an NGO coordinator, I want the system to automatically recommend suitable volunteers for a task using AI, so that I can assign the right people without manual searching.
- US-05 — As a volunteer, I want to receive AI-matched task recommendations based on my profile, so that I can select activities aligned with my skills and interests.
- US-06 — As an NGO administrator, I want the AI engine to learn from past volunteer performance ratings, so that future matching recommendations improve over time.

EPIC-3: Project & Task Progress Monitoring

- US-07 — As an NGO coordinator, I want to create projects and break them into tasks with deadlines, so that work is organized and trackable.
- US-08 — As a volunteer, I want to update the status of my assigned task in real time, so that coordinators have visibility into progress.
- US-09 — As an NGO administrator, I want a real-time dashboard showing project completion percentage, so that I can identify delays and reallocate resources.

EPIC-4: Donation & Resource Tracking

- US-10 — As a donor, I want to make an online donation towards a specific project, so that I can contribute conveniently and securely.
- US-11 — As an NGO administrator, I want to record and categorize incoming donations (cash, kind, resources), so that all contributions are accurately tracked.
- US-12 — As a donor, I want to view a report on how my donation was utilized, so that I can trust the NGO's transparency.

EPIC-5: Reporting & Analytics Dashboard

- US-13 — As an NGO administrator, I want to generate monthly reports on volunteer participation and project outcomes, so that I can share progress with stakeholders and donors.
- US-14 — As a government regulator, I want to access compliance reports on fund utilization, so that I can verify the NGO's adherence to regulations.
- US-15 — As an NGO administrator, I want visual analytics (charts and graphs) of key performance indicators, so that I can make informed decisions quickly.

EPIC-6: Communication & Notification System

- US-16 — As a volunteer, I want to receive push notifications about new task assignments and reminders, so that I never miss an activity.
- US-17 — As an NGO coordinator, I want to broadcast announcements to all volunteers in a project, so that everyone stays informed of updates.
- US-18 — As a donor, I want to receive notifications when a project I funded reaches a milestone, so that I stay engaged with its progress.

Figure 4: High-Level System Architecture

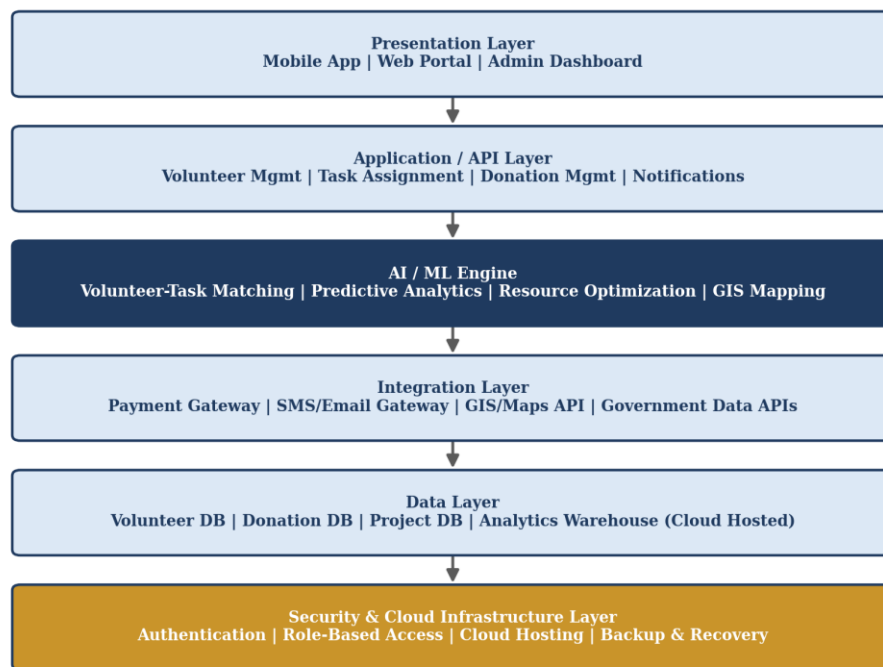


Figure 2: High-Level System Architecture Supporting the Epics

4. Acceptance Criteria (Given–When–Then)

Acceptance criteria define the specific, testable conditions that must be satisfied for a user story to be considered complete. The Given–When–Then (GWT) format was used to express these criteria clearly, aligning development and QA teams on the expected system behavior. The ten highest-priority user stories from the backlog are detailed below; remaining stories follow an equivalent structure during backlog refinement.

US-01: Volunteer Profile Creation

Given a new user has downloaded the platform and selected “Register as Volunteer”

When they enter personal details, skills, interests, and availability, and submit the form

Then a volunteer profile is created and stored, and a confirmation notification is sent to the user.

US-04: AI Recommends Volunteers for a Task

Given a coordinator has created a new task with required skills and location

When the coordinator selects “Find Volunteers”

Then the AI engine displays a ranked list of the top matching, available volunteers within 5 seconds.

US-05: Volunteer Receives AI-Matched Recommendations

Given a volunteer has an active profile with defined skills and availability

When new tasks matching their profile are published on the platform

Then the volunteer receives a personalized notification listing recommended tasks ranked by match score.

5. Product Backlog Prioritization

The product backlog was prioritized using a combination of business value scoring (on a scale of 1–10) and the MoSCoW technique (Must have, Should have, Could have, Won't have this time), while explicitly accounting for technical dependencies between stories. Stories that unlock core end-to-end workflows — profile creation, project setup, AI matching, and donations — were rated highest, since they deliver the foundational value the remaining backlog depends upon. Compliance and advanced-analytics stories, while important, were prioritized lower for the first release as they build upon data generated by the core workflows.

ID	Epic	User Story (Summary)	Business Value	Priority	Dependency
US-01	EPIC-1	Volunteer creates digital profile	9	Must	None
US-02	EPIC-1	Volunteer uploads verification documents	6	Should	US-01
US-03	EPIC-1	Administrator approves registration	8	Must	US-01
US-04	EPIC-2	AI recommends volunteers for a task	10	Must	US-01, US-07
US-05	EPIC-2	Volunteer receives AI-matched recommendations	9	Must	US-01
US-06	EPIC-2	AI learns from performance ratings	7	Should	US-04, US-05

ID	Epic	User Story (Summary)	Business Value	Priority	Dependency
US-07	EPIC-3	Coordinator creates projects and tasks	10	Must	None
US-08	EPIC-3	Volunteer updates task status	9	Must	US-07
US-09	EPIC-3	Real-time project progress dashboard	9	Must	US-07, US-08
US-10	EPIC-4	Donor makes online donation	10	Must	None
US-11	EPIC-4	Administrator records/categorizes donations	8	Must	US-10
US-12	EPIC-4	Donor views utilization report	7	Should	US-11
US-13	EPIC-5	Administrator generates monthly reports	6	Should	US-09, US-11
US-14	EPIC-5	Regulator accesses compliance reports	6	Could	US-11
US-15	EPIC-5	Administrator views visual analytics	6	Should	US-09
US-16	EPIC-6	Volunteer receives push notifications	8	Must	US-04
US-17	EPIC-6	Coordinator broadcasts announcements	5	Could	US-07
US-18	EPIC-6	Donor receives milestone notifications	5	Could	US-09, US-10
US-19	EPIC-7	Immutable audit trail for transactions	8	Must	US-10, US-11
US-20	EPIC-7	Role-based access control	9	Must	US-01, US-03
US-21	EPIC-7	Regulator exports compliance reports	5	Won't (this release)	US-14

This prioritization ensures that Sprint 1 focuses on delivering a working, demonstrable slice of the platform's core value proposition — digitized registration, AI-assisted matching, project tracking, and secure donations — while deferring dependent reporting and compliance features to subsequent sprints once underlying data flows are in place.

Figure 2: AI-Powered Volunteer-Task Matching Workflow

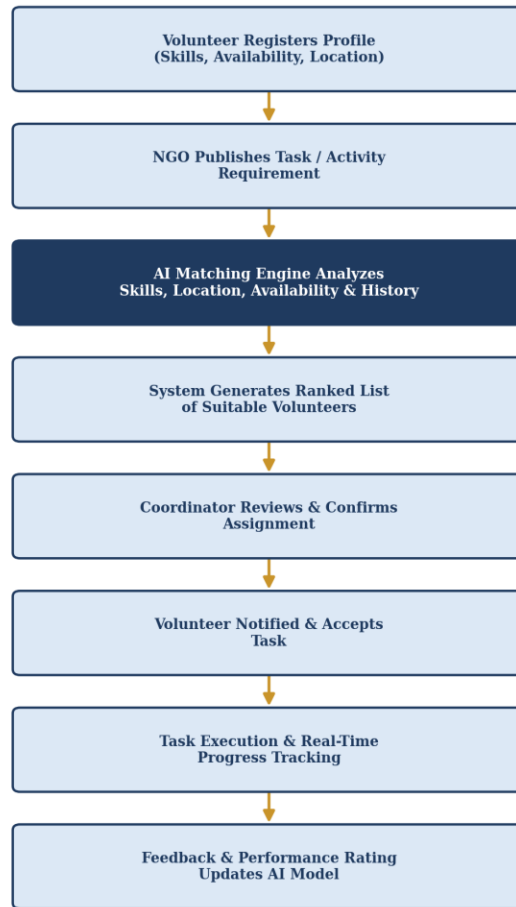


Figure 3: AI-Powered Volunteer-Task Matching Workflow (a Must-Have, Highest-Value Capability)

6. Sprint Backlog Planning (Sprint 1)

Based on business value, dependency structure, and the team's estimated velocity, eight high-priority user stories were selected for Sprint 1 (a 2-week sprint). These stories collectively deliver an end-to-end demonstrable workflow: volunteers can register, NGOs can create and assign projects, the AI engine can perform baseline matching, task progress can be tracked in real time, and donors can make secure, tracked donations. Stories US-03, US-11, US-16, and US-19 are carried into Sprint 2, as they depend on data and services first established in Sprint 1.

Sprint 1 Selected Stories

ID	Selected User Story
US-01	As a volunteer, I want to create a digital profile with my skills, interests, and availability, so that NGOs can identify me for suitable activities.
US-07	As an NGO coordinator, I want to create projects and break them into tasks with deadlines, so that work is organized and trackable.
US-04	As an NGO coordinator, I want the system to automatically recommend suitable volunteers for a task using AI, so that I can assign the right people without manual searching.
US-05	As a volunteer, I want to receive AI-matched task recommendations based on my profile, so that I can select activities aligned with my skills and interests.
US-08	As a volunteer, I want to update the status of my assigned task in real time, so that coordinators have visibility into progress.
US-09	As an NGO administrator, I want a real-time dashboard showing project completion percentage, so that I can identify delays and reallocate resources.
US-10	As a donor, I want to make an online donation towards a specific project, so that I can contribute conveniently and securely.
US-20	As an NGO administrator, I want role-based access control for sensitive data, so that only authorized personnel can view or modify financial and volunteer records.

Sprint 1 Task Breakdown

Each selected user story was decomposed into concrete, assignable development tasks to guide daily execution during the sprint.

Story ID	Task	Owner Role	Est. Hours
US-01	Design registration UI/UX wireframes	UI/UX Designer	6
	Develop registration API & database schema	Backend Developer	10
	Build registration form (frontend)	Frontend Developer	8
	Write unit & integration tests	QA Engineer	5
US-07	Design project/task data model	Backend Developer	6
	Develop project & task creation API	Backend Developer	8
	Build admin project-creation UI	Frontend Developer	8
	Test project/task CRUD operations	QA Engineer	4

Story ID	Task	Owner Role	Est. Hours
US-04	Define matching feature set (skills, location, availability)	Data Scientist	8
	Train baseline AI matching model	Data Scientist	12
	Integrate matching API with backend	Backend Developer	8
	Build “Find Volunteers” UI component	Frontend Developer	6
US-05	Develop recommendation notification service	Backend Developer	6
	Build volunteer recommendation feed UI	Frontend Developer	6
US-08	Develop task status update API & real-time sync	Backend Developer	8
	Build mobile task-status update screen	Mobile Developer	8
US-09	Design dashboard wireframes	UI/UX Designer	5
	Develop dashboard aggregation API	Backend Developer	10
	Build dashboard charts & widgets	Frontend Developer	10
	Load-test dashboard performance	QA Engineer	4
US-10	Integrate payment gateway	Backend Developer	10
	Build donation checkout UI	Frontend Developer	8
	Implement receipt generation	Backend Developer	4
	Security-test the payment flow	QA / Security Engineer	6
US-20	Design role-based access control (RBAC) model	Backend Developer	6
	Implement RBAC middleware	Backend Developer	8
	Configure permissions for admin/coordinator/volunteer roles	Backend Developer	4
	Test access-control scenarios	QA Engineer	4

Figure 3: Agile Sprint Workflow Cycle

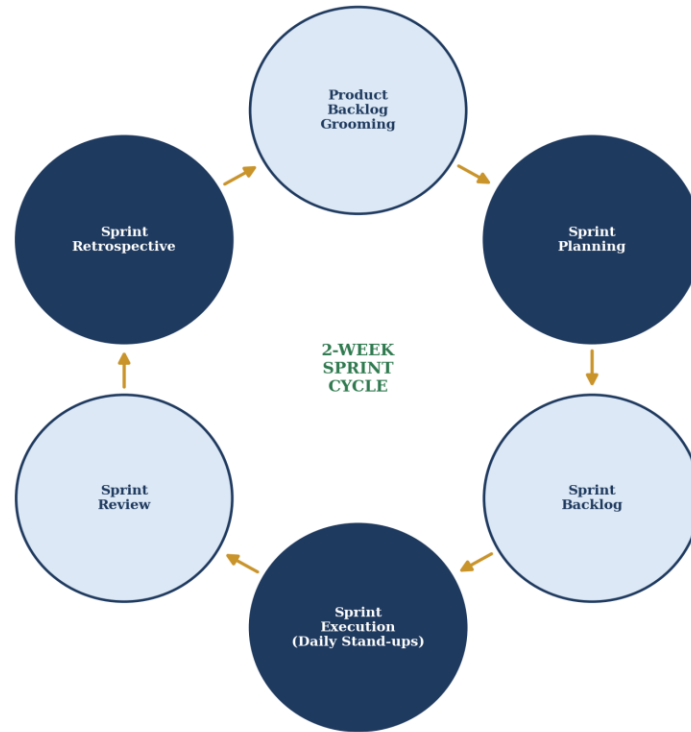


Figure 4: Agile Sprint Workflow Cycle Followed for Sprint 1

7. Story Point Estimation

Story points were estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13), a technique that reduces anchoring bias by having each cross-functional team member (developers, QA engineers, a data scientist, and a UI/UX designer) estimate independently before discussing and converging on a consensus value. Estimates reflect a combination of technical complexity, uncertainty, and effort relative to a previously agreed baseline story.

Sprint 1 commits a total of 57 story points across the eight selected stories. Based on the composition of the cross-functional team (approximately six to eight members, including backend, frontend, mobile, data science, QA, and design roles) and an assumed baseline velocity derived from comparable past sprints, this workload is considered an ambitious but achievable commitment for a two-week sprint, with US-04 (AI matching) treated as the sprint's highest-risk item and monitored closely during daily stand-ups.

8. Sprint Goal and Expected Deliverables

Sprint Goal

Enable the core end-to-end workflow of the platform: volunteers can register and build profiles, NGO administrators can create projects and tasks, the AI engine can perform an initial round of volunteer-task matching, task progress can be tracked in real time, and donors can make their first securely tracked online donations — establishing the foundational backbone of the platform ahead of iterative enhancement in subsequent sprints.

Definition of Done

A user story is considered done when its acceptance criteria are met and verified by QA, the code has been peer-reviewed and merged, automated tests pass in the continuous integration pipeline, and the feature has been successfully demonstrated in the staging environment during the Sprint Review.

9. Conclusion

This sprint planning document applies core Agile software engineering practices — vision framing, epic and user-story decomposition, Given–When–Then acceptance criteria, backlog prioritization, sprint planning, Planning Poker estimation, and clear sprint-goal articulation — to the AI-Powered Smart Volunteer and NGO Resource Management Platform. By focusing Sprint 1 on the highest business-value, foundational capabilities (registration, project creation, AI-based matching, progress tracking, and donations), the team establishes a working, demonstrable core of the platform within the first iteration.

Subsequent sprints will build upon this foundation to deliver deferred capabilities — including detailed donation-utilization reporting, compliance exports, advanced analytics, and expanded communication features — progressively increasing the platform's ability to improve volunteer engagement, resource efficiency, and transparency across the NGO's operations, ultimately maximizing its social impact.